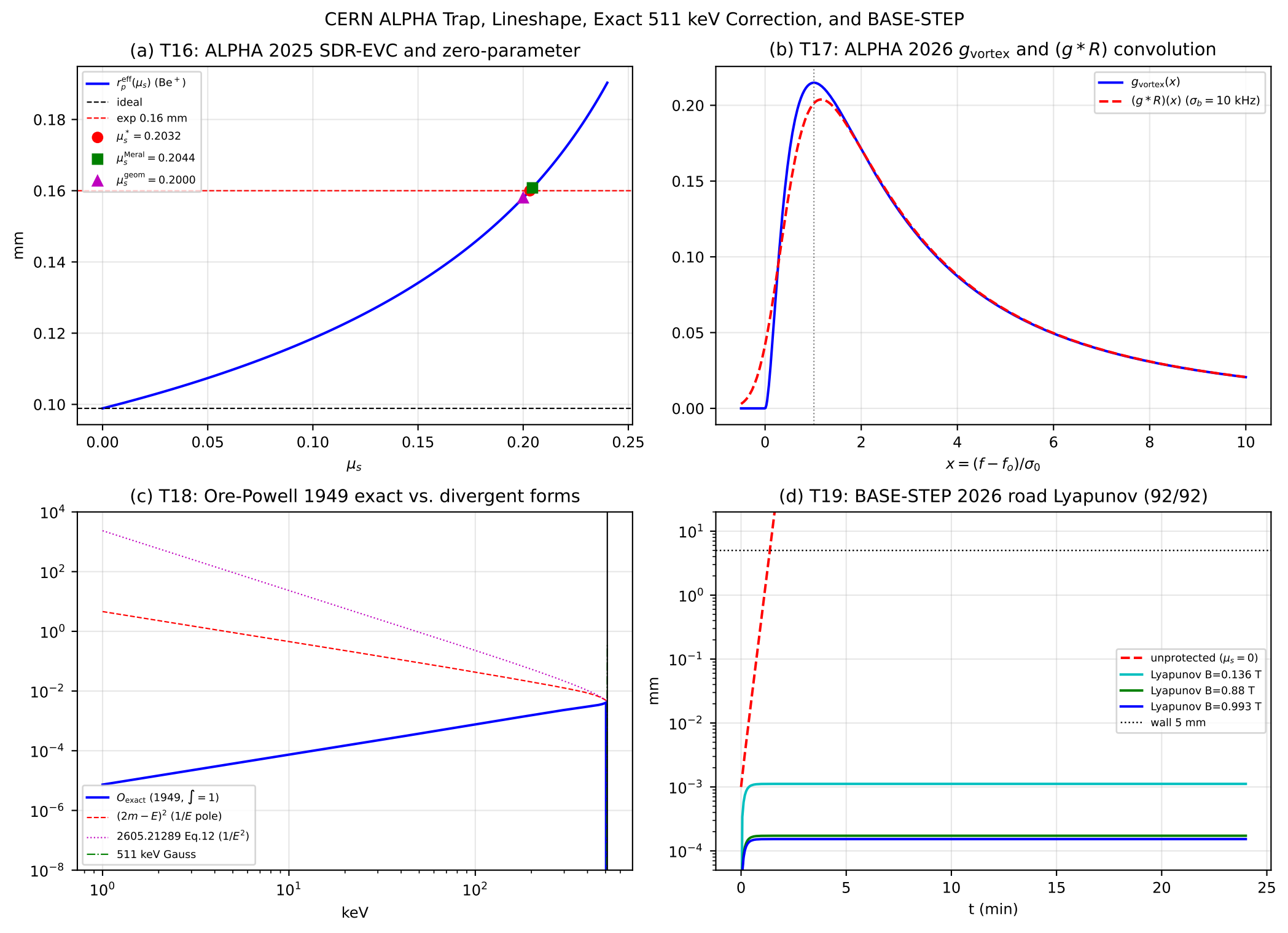
CERN ALPHA Trap, Lineshape, Exact 511 keV Correction, and BASE-STEP
(a) T16: ALPHA 2025 SDR-EVC and zero-parameter
$r_p^{\mathrm{eff}}(\mu_s)$ (Be$^+$)
ideal
exp 0.16 mm
$\mu_s^* = 0.2032$
$\mu_s^{\mathrm{Meral}} = 0.2044$
$\mu_s^{\mathrm{geom}} = 0.2000$
mm
μ_s
(b) T17: ALPHA 2026 g_{vortex} and $(g * R)$ convolution
$g_{\mathrm{vortex}}(x)$
$(g * R)(x)$ ($\sigma_b = 10$ kHz)
$x = (f - f_o)/\sigma_0$
(c) T18: Ore-Powell 1949 exact vs. divergent forms
O_{exact} (1949, $\int = 1$)
$(2m - E)^2$ (1/E pole)
2605.21289 Eq.12 ($1/E^2$)
511 keV Gauss
keV
(d) T19: BASE-STEP 2026 road Lyapunov (92/92)
unprotected ($\mu_s = 0$)
Lyapunov B=0.136 T
Lyapunov B=0.88 T
Lyapunov B=0.993 T
wall 5 mm
mm
t (min)